

Functional Annotation Report: *acnA-I* (PP_2112, UniProt Q88L24)

Aconitate hydratase (Aconitase, EC 4.2.1.3) — *Pseudomonas putida* KT2440

1. Summary (Answer to the Research Question)

acnA-I (PP_2112, Q88L24) encodes a cytoplasmic, [4Fe-4S]-cluster-dependent aconitase (aconitate hydratase, EC 4.2.1.3) that catalyzes the reversible, stereospecific isomerization of citrate to *D-threo*-isocitrate via the enzyme-bound intermediate *cis*-aconitate — the second step of the tricarboxylic acid (TCA/Krebs) cycle. Sequence, structural, and evolutionary evidence identify it specifically as an **AcnA-type** enzyme (73% identical to the biochemically characterized *Escherichia coli* AcnA), i.e. the stable, iron/oxidative-stress-inducible "maintenance" aconitase, as opposed to the exponential-phase catabolic AcnB or the propionate/2-methylcitrate-dedicated aconitases encoded elsewhere in the genome. Like other AcnA/IRP-family proteins, it is predicted to be **bifunctional**: when its Fe-S cluster is lost under iron limitation or oxidative stress, the apo-protein can act as a post-transcriptional mRNA-binding regulator.

Gene identity was verified at every step: the symbol *acnA-I*, the aconitate hydratase description, the aconitase/IPM-isomerase family, the AcnA_IRP swivel and aconitase domains, and the organism *P. putida* KT2440 are all mutually consistent with the enzyme characterized below. There was **no** evidence of a same-symbol collision with an unrelated gene.

2. Molecular Function and Reaction Catalyzed

2.1 The reaction

AcnA-I is an **aconitate hydratase (aconitase)**, EC 4.2.1.3, catalyzing:



The overall reaction is a net isomerization (a dehydration followed by a stereochemically distinct re-hydration) that moves a hydroxyl group from the C3 to the C2 position of the tricarboxylic acid, proceeding through the tightly bound intermediate *cis*-aconitate (UniProt Q88L24 catalytic activity: "citrate = D-threo-isocitrate"; UniPathway: *tricarboxylic acid cycle; isocitrate from oxaloacetate: step 2/2*). Aconitases are iron–sulfur proteins that "catalyse the reversible isomerization of citrate and isocitrate via the intermediate *cis*-aconitate in the Krebs cycle" (Baothman et al., 2013,).

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2.2 Catalytic cofactor and mechanism

Catalysis depends on a **[4Fe-4S] cluster** (UniProt cofactor annotation; keywords 4Fe-4S, Iron-sulfur). Three of the four irons are ligated by cysteine residues; the fourth iron (**Fe4**) is coordinatively unsaturated and directly binds the substrate's hydroxyl and a carboxylate, acting as a Lewis acid during the dehydration/rehydration. Crystallographic studies of the homologous mitochondrial aconitase established that "Fe4 of the [4Fe-4S] cluster is five-coordinate," and that the mechanistic model extends "to all three natural substrates of aconitase" — citrate, *cis*-aconitate, and isocitrate (Lauble et al., 1994,).

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Residue-level evidence for an intact active site in AcnA-I: - The three cluster-ligating cysteines occur in the canonical single-Cys + Cys-X-X-Cys geometry: **Cys454**, **Cys520**, **Cys523** in PP_2112. - These map **one-to-one** onto the curated *E. coli* AcnA ligands **Cys435**, **Cys501**, **Cys504** (UniProt P25516), whose Fe-S cluster was characterized by EPR and magnetic-CD spectroscopy and shown to "strongly resemble those of the mammalian enzymes" (Jordan et al., 1999,).

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- The active-site stretch **ITSCTNTSNP** (surrounding the first ligand cysteine) is **identical** in PP_2112 (452–461) and *E. coli* AcnA (433–442), indicating full conservation of the catalytic pocket.

2.3 Substrate specificity

AcnA-I is expected to act on the **canonical TCA tricarboxylates (citrate/*cis*-aconitate/isocitrate)**. In some AcnA enzymes the same active site can also hydrate 2-methyl-*cis*-aconitate to 2-methylisocitrate in the 2-methylcitrate (propionate) cycle (Horswill & Escalante-Semerena, 2001,

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However, in *P. putida* KT2440 that propionate-catabolic role is delegated to a **separate, dedicated gene cluster** (see Section 4), so AcnA-I's physiological substrate is the standard TCA-cycle citrate/isocitrate pair.

3. Subcellular Localization

AcnA-I functions in the **cytoplasm (cytosol)**. UniProt Q88L24 annotates only structural domains (aconitase large subunit, residues 76–581; AcnA/IPM swivel, 710–837) and contains **no signal peptide and no transmembrane segment**, ruling out periplasmic, membrane, or extracellular localization. This is consistent with the aconitase/IRP1 family being soluble cytosolic proteins — mammalian IRP1/cytosolic aconitase is "a well-established cytosolic protein" (Gu et al., 2022,

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The TCA cycle and the enzyme's potential mRNA targets both reside in the same compartment.

4. Genomic Context, Paralogs, and Pathway Placement

P. putida KT2440 encodes **three** aconitase-family (EC 4.2.1.3) proteins plus an aconitate isomerase, revealing a division of labour:

Locus	Gene	UniProt	Length	Role (inferred)
PP_2112	acnA-I	Q88L24	913 aa	Standalone AcnA-type TCA-cycle aconitase (this report)
PP_2336	acnA-II	Q88KF4	862 aa	AcnA-type, within the propionate/2-methylcitrate cluster
PP_2337	prpF	Q88KF3	396 aa	Aconitate isomerase (2-methyl-aconitate handling)
PP_2339	acnB	Q88KF1	869 aa	AcnB-type; also 2-methylisocitrate dehydratase (EC 4.2.1.99)

acnA-I (PP_2112) sits alone on the chromosome, whereas acnA-II/prpF/acnB form a contiguous cluster associated with **2-methylcitrate / propionate catabolism**. This mirrors the observation that acnA-type genes recur "in a number of putative prp operons" (Horswill & Escalante-Semerena, 2001,

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— but importantly, PP_2112 is the copy that is *not* in such an operon, marking it as the **general TCA-cycle housekeeping/maintenance aconitase**.

Pathway: The reaction is step 2 of the TCA cycle (citrate → isocitrate), feeding isocitrate to isocitrate dehydrogenase and, at branch points, to the glyoxylate shunt. In *P. putida*, whose metabolism is organised around the TCA cycle and gluconeogenesis, this step is central to energy generation and to biosynthetic precursor supply (2-oxoglutarate, succinyl-CoA, oxaloacetate).

5. Evolutionary / Bioinformatic Evidence for the AcnA Assignment

Global (Needleman–Wunsch) pairwise identity of PP_2112 against reference aconitases:

Reference	UniProt	% identity to PP_2112
<i>E. coli</i> AcnA	P25516	72.6%
<i>P. putida</i> AcnA-II	Q88KF4	54.3%
Pig mitochondrial aconitase / IRP1 homolog	P16276	43.3%
<i>P. putida</i> AcnB	Q88KF1	34.5%
<i>E. coli</i> AcnB	P36683	34.9%

The ~2-fold identity gap between the AcnA clade (>54%) and the AcnB clade (~35%) unambiguously assigns PP_2112 to the **AcnA subfamily**. Because *E. coli* AcnA is biochemically well characterized, this **orthology** (not merely family membership) licenses transfer of its established properties to PP_2112.

Independent structural support: the AlphaFold model of AcnA-I (AF-Q88L24-F1) is **very high confidence** (mean pLDDT 92.4; 96.8% of residues >70, 80.4% >90), indicating the four-domain aconitase architecture (three large-subunit domains plus the AcnA/IPM swivel domain closing over the active-site cleft) is modeled reliably — consistent with a canonical aconitase fold competent for [4Fe-4S] binding and catalysis.

Properties of the AcnA archetype (transferable to AcnA-I): - A **stable, aerobic stationary-phase enzyme, specifically induced by iron and oxidative stress**, with higher affinity for citrate and a broad pH optimum — "consistent with its role as a maintenance or survival enzyme during nutritional or oxidative stress," in contrast to the catabolic, less stable AcnB (Jordan et al., 1999,).

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- In *Salmonella*, acnA expression is controlled by CRP (glucose starvation), FNR (oxygen starvation), Fur (iron starvation), and SoxR (oxidative stress) (Baothman et al., 2013,).

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6. Secondary (Moonlighting) Function: Cluster-Dependent RNA Binding

AcnA-family proteins are **bifunctional**, resembling the vertebrate iron-regulatory proteins (IRP1) in "having alternative mRNA-binding and catalytic activities" (Tang & Guest, 1999,).

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When the labile [4Fe-4S] cluster is disassembled by oxidants or iron starvation, the **apo-protein** switches to a site-specific RNA-binding mode: - In *E. coli*, apo-AcnA/AcnB bind the 3'UTRs of acn mRNAs, and the apo-forms have >3-fold higher mRNA affinity than the holo-enzymes; "the apo-proteins function as post-transcriptional regulators by site-specific binding to acn mRNAs," with AcnA **stabilizing sodA (superoxide dismutase) mRNA** — coupling aconitase status to the oxidative-stress defence (Tang et al., 2002,).

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- The switch is cluster-dependent, exactly as for mammalian IRP1, which is "a bifunctional protein with mutually exclusive RNA-binding or enzymatic activities that depend on the presence of a 4Fe-4S cluster" (Gu et al., 2022,).

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PP_2112 carries the UniProt **RNA-binding** keyword (GO:0003723) and the **AcnA_IRP swivel domain (IPR044137)**, so this regulatory potential is structurally encoded. Direct experimental demonstration of RNA targets specifically for the *P. putida* protein has not been reported; this role is inferred from strong orthology and conserved domain architecture.

7. Supported and Refuted Hypotheses

Supported: - H1 — AcnA-I is a genuine [4Fe-4S] aconitase catalyzing citrate↔isocitrate in the TCA cycle. (*Curated annotation + residue-level active-site conservation.*) - H2 — PP_2112 is an AcnA-type (not AcnB-type) enzyme, orthologous to *E. coli* AcnA. (*73% vs 35% identity; clade separation.*) - H3 — The enzyme is cytoplasmic. (*No signal/TM; family is cytosolic.*) - H4 — Physiological substrate is the standard TCA citrate/isocitrate pair, with propionate/2-methylcitrate handled by the separate acnA-II/acnB/prpF cluster. (*Paralog/genomic-context analysis.*)

Supported by inference (not yet experimentally confirmed in *P. putida*): - H5 — Apo-AcnA-I can moonlight as a post-transcriptional RNA-binding regulator under iron/oxidative stress. (*Orthology + IRP swivel domain + RNA-binding keyword; awaits direct experimental validation.*)

Refuted: - That PP_2112 is the exponential-phase catabolic AcnB (35% identity — rejected) or a dedicated 2-methylcitrate-cycle enzyme (that function maps to PP_2336–2339 — rejected as its primary role).

8. Limitations and Future Directions

- **Direct experimental data specific to PP_2112 are sparse.** Targeted PubMed searches returned no primary *P. putida* AcnA-I characterization; conclusions rest on (i) curated database annotation, (ii) very high orthology to the experimentally characterized *E. coli*/*Salmonella* AcnA, and (iii) residue-level structural conservation. This is strong but indirect for the moonlighting role.
- **Future work:** purify AcnA-I and measure kinetic constants (K_m for citrate/isocitrate) and pH profile; test single/double aconitase knockouts (*acnA-I* vs *acnA-II/acnB*) for growth-phase, iron-, and oxidative-stress phenotypes; perform EPR/MCD on the cluster; and use RNA-immunoprecipitation/CLIP to identify apo-AcnA-I mRNA targets in *P. putida*.

9. Key References

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- UniProt Q88L24 (acnA-I / PP_2112); P25516 (*E. coli* AcnA); P36683 (*E. coli* AcnB); Q88KF1/Q88KF4/Q88KF3 (*P. putida* paralogs).